

## PRINCE RUPERT, BC, Canada

WMO: 710220

Lat: 54.290N

Lon: 130.440W

Elev: 116

StdP: 14.63

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 16.3       | 20.7 | 0.5                           | 5.7 | 22.1 | 8.0 | 8.3 | 24.3 | 31.8                  | 44.9 | 28.6 | 44.0 | 5.3                   | 70   | 0.635 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.9                    | 67.2            | 58.8 | 64.5 | 57.9 | 62.5 | 57.1 | 60.6                | 64.8 | 59.3 | 62.8 | 58.2 | 61.2 | 7.1                  | 290  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 58.6                            | 73.7 | 61.9 | 57.5 | 70.9 | 60.8 | 56.6 | 68.4 | 59.8 | 26.8          | 65.1 | 25.9 | 63.0 | 25.2 | 61.3 | 67.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               | Max  |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 25.1              | 21.1 | 18.0 | 11.0                       | 75.2 | 6.5                | 4.7 | 6.3                                                | 78.6 | 2.5        | 81.4 | -1.2       | 84.0 | -5.9       | 87.4 |
|                   |      |      | 9.6                        | 63.6 | 6.3                | 1.6 | 5.1                                                | 64.8 | 1.4        | 65.8 | -2.2       | 66.7 | -6.8       | 67.9 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.2   | 37.1  | 37.3  | 39.7  | 43.9  | 49.2  | 53.6  | 56.8  | 57.5  | 53.9  | 47.5  | 40.7  | 37.1 |
|                                                                                  | DBStd       | 8.96   | 7.52  | 5.82  | 5.09  | 3.68  | 3.92  | 3.24  | 2.72  | 2.63  | 3.44  | 3.98  | 5.91  | 6.73 |
|                                                                                  | HDD50       | 2110   | 400   | 355   | 321   | 186   | 62    | 4     | 0     | 0     | 7     | 96    | 280   | 399  |
|                                                                                  | HDD65       | 6851   | 865   | 775   | 786   | 632   | 491   | 343   | 254   | 234   | 334   | 544   | 729   | 864  |
|                                                                                  | CDD50       | 736    | 0     | 0     | 0     | 4     | 36    | 111   | 211   | 232   | 123   | 17    | 2     | 0    |
|                                                                                  | CDD65       | 2      | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 17     | 0     | 0     | 0     | 0     | 1     | 11    | 1     | 3     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 2      | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.5    | 8.3   | 8.0   | 8.7   | 8.0   | 6.9   | 6.3   | 5.6   | 5.7   | 6.7   | 8.3   | 8.7   | 8.9  |
| Precipitation                                                                    | PrecAvg     | 105.9  | 10.2  | 8.8   | 7.6   | 7.2   | 5.7   | 4.8   | 4.6   | 6.7   | 10.0  | 14.8  | 11.8  | 11.5 |
|                                                                                  | PrecMax     | 129.2  | 17.2  | 15.7  | 12.3  | 13.3  | 13.2  | 9.5   | 9.4   | 21.3  | 20.3  | 29.4  | 23.7  | 22.3 |
|                                                                                  | PrecMin     | 90.4   | 3.8   | 0.2   | 1.5   | 2.4   | 2.4   | 1.2   | 0.6   | 1.2   | 3.1   | 6.4   | 3.7   | 3.8  |
|                                                                                  | PrecStd     | 10.4   | 3.2   | 3.7   | 2.5   | 2.8   | 2.5   | 2.1   | 2.4   | 3.6   | 4.3   | 4.6   | 4.5   | 4.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.3  | 52.1  | 57.4  | 62.7  | 69.6  | 73.2  | 70.6  | 71.7  | 68.3  | 60.6  | 54.8  | 52.8 |
|                                                                                  |             | MCWB   | 46.3  | 44.1  | 47.6  | 48.6  | 53.7  | 60.2  | 61.9  | 60.8  | 59.9  | 52.4  | 48.0  | 45.7 |
|                                                                                  | 2%          | DB     | 49.0  | 49.5  | 51.2  | 56.3  | 61.6  | 63.8  | 65.2  | 66.2  | 63.7  | 57.2  | 51.9  | 49.4 |
|                                                                                  |             | MCWB   | 45.7  | 43.7  | 44.6  | 47.0  | 52.5  | 56.8  | 59.3  | 59.6  | 57.1  | 52.0  | 48.0  | 45.0 |
|                                                                                  | 5%          | DB     | 47.3  | 47.2  | 48.5  | 52.4  | 57.6  | 60.8  | 63.1  | 64.1  | 61.1  | 54.9  | 50.1  | 47.5 |
|                                                                                  |             | MCWB   | 44.5  | 42.8  | 43.7  | 46.1  | 51.6  | 55.6  | 58.1  | 58.7  | 56.0  | 50.7  | 47.3  | 44.0 |
|                                                                                  | 10%         | DB     | 45.8  | 45.0  | 46.4  | 50.1  | 55.2  | 58.6  | 61.5  | 62.1  | 59.1  | 53.2  | 48.4  | 45.4 |
|                                                                                  |             | MCWB   | 43.2  | 42.0  | 42.5  | 45.1  | 50.5  | 54.2  | 57.2  | 57.8  | 55.2  | 49.8  | 46.0  | 42.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 48.3  | 47.7  | 48.4  | 52.3  | 56.0  | 60.2  | 62.2  | 62.3  | 61.2  | 56.1  | 50.8  | 48.0 |
|                                                                                  |             | MCDB   | 49.8  | 49.6  | 53.6  | 57.8  | 63.6  | 67.6  | 68.5  | 68.1  | 65.5  | 58.2  | 52.8  | 50.1 |
|                                                                                  | 2%          | WB     | 46.5  | 45.0  | 46.2  | 48.9  | 53.9  | 57.8  | 60.2  | 60.7  | 58.6  | 53.6  | 49.2  | 46.2 |
|                                                                                  |             | MCDB   | 48.3  | 47.2  | 49.6  | 52.8  | 58.6  | 62.7  | 64.2  | 64.8  | 61.7  | 55.3  | 51.0  | 48.5 |
|                                                                                  | 5%          | WB     | 44.9  | 43.6  | 44.7  | 47.2  | 52.5  | 56.3  | 58.9  | 59.5  | 57.2  | 51.7  | 47.7  | 44.4 |
|                                                                                  |             | MCDB   | 46.7  | 46.1  | 47.5  | 51.0  | 56.2  | 59.9  | 62.1  | 62.8  | 59.7  | 53.4  | 49.7  | 46.7 |
|                                                                                  | 10%         | WB     | 43.5  | 42.2  | 43.2  | 45.7  | 51.1  | 54.9  | 57.8  | 58.4  | 55.9  | 50.4  | 46.1  | 42.9 |
|                                                                                  |             | MCDB   | 45.3  | 44.8  | 45.8  | 49.4  | 54.4  | 57.8  | 60.6  | 61.3  | 58.3  | 52.5  | 48.2  | 45.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.9   | 9.8   | 10.0  | 10.9  | 10.3  | 9.0   | 8.2   | 8.9   | 9.7   | 9.1   | 8.5   | 8.0  |
|                                                                                  |             | MCDBR  | 8.7   | 11.7  | 12.9  | 15.9  | 15.2  | 13.3  | 11.3  | 12.7  | 13.1  | 11.4  | 9.2   | 8.6  |
|                                                                                  |             | MCWBR  | 6.3   | 7.6   | 8.0   | 9.1   | 8.3   | 7.1   | 6.4   | 7.1   | 7.7   | 7.4   | 6.7   | 5.9  |
|                                                                                  | 5% WB       | MCDBR  | 7.6   | 9.6   | 11.1  | 13.2  | 12.9  | 11.8  | 10.5  | 11.3  | 10.8  | 8.7   | 8.2   | 7.8  |
|                                                                                  |             | MCWBR  | 6.2   | 7.0   | 7.5   | 8.3   | 7.7   | 6.8   | 6.4   | 6.9   | 7.1   | 6.5   | 6.7   | 6.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.301  | 0.308 | 0.334 | 0.369 | 0.360 | 0.380 | 0.366 | 0.363 | 0.355 | 0.336 | 0.310 | 0.290 |      |
|                                                                                  | taud        | 2.374  | 2.483 | 2.421 | 2.331 | 2.435 | 2.404 | 2.489 | 2.519 | 2.515 | 2.476 | 2.354 | 2.313 |      |
|                                                                                  | Ebn at Noon | 209    | 253   | 269   | 272   | 281   | 276   | 278   | 272   | 259   | 237   | 200   | 184   |      |
|                                                                                  | Edh at Noon | 17     | 22    | 28    | 35    | 33    | 35    | 32    | 29    | 26    | 22    | 17    | 15    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 163    | 421   | 753   | 1207  | 1548  | 1633  | 1483  | 1281  | 844   | 463   | 215   | 131   |      |
|                                                                                  | RadStd      | 21     | 74    | 95    | 75    | 144   | 124   | 169   | 113   | 96    | 68    | 30    | 23    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (18 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page